

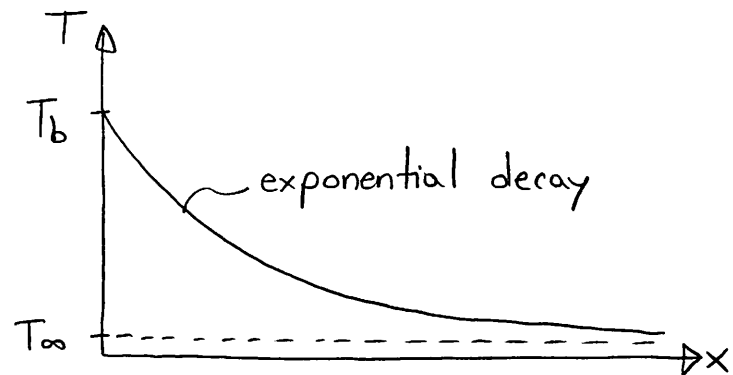
Applying our first B.C.:

$$\theta(x=0) = \theta_b = C_2 \underbrace{e^{-m(0)}}_{e^0=1} \Rightarrow C_2 = \theta_b$$

$$\boxed{\theta(x) = \theta_b e^{-mx}}$$

Or in dimensional form:

$$\boxed{\frac{T - T_\infty}{T_b - T_\infty} = e^{-\sqrt{\frac{hP'}{kA}} \cdot x}} \Rightarrow \text{Fin temperature profile:}$$



So how do we calculate heat transfer?

Let's calculate it at the base because all of our energy comes from the base:

$$Q_{\text{net}} = Q(x=0) = -kA \left. \frac{\partial \theta}{\partial x} \right|_{x=0} = +kA \theta_b \underbrace{e^{-m(0)}}_1$$

$$\boxed{Q_{\text{net}} = q_{\text{net}} = +kA \theta_b \sqrt{\frac{hP'}{kA}} = \sqrt{kAhP'} \theta_b}$$

Note, if we compare our heat transfer to that of the case with no fin, we get:

$$\frac{Q_{\text{net}}}{Q_0} = \frac{\sqrt{kAhP'} \theta_b}{hA \theta_b} = \sqrt{\frac{kP'}{hA}} = \sqrt{\frac{k}{ht}} = Bi_t^{-1/2} \gg 1 \text{ since } Bi_t \gg 1$$

Hence adding the fin enhances the heat transfer very much!

Note here I assumed: $\frac{P}{A} = \frac{1}{L}$

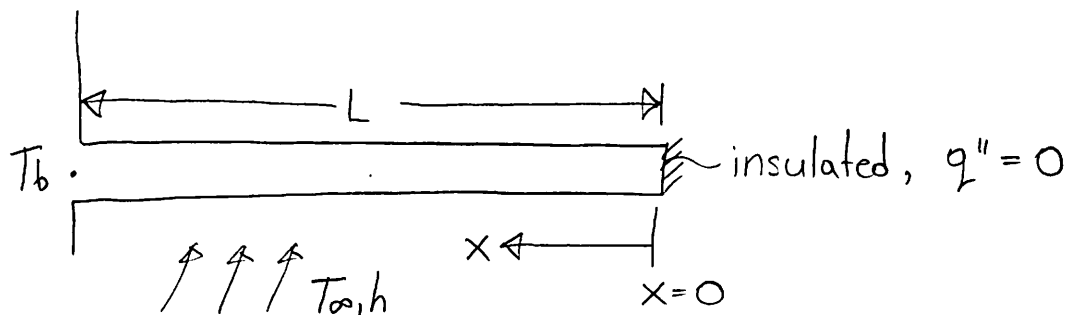
We can do another sanity check:

$$Q_{\text{net}} = \int_0^{\infty} hP\theta dx = \int_0^{\infty} hP\theta_b e^{-mx} dx$$

$$= hP\theta_b \frac{e^{-mx}}{m} \Big|_0^{\infty} = \frac{hP\theta_b}{m} = \sqrt{kAhP}\theta_b$$

Works!
Same answer as before!

② Insulated Tip $\left(-k \frac{\partial T}{\partial x} \Big|_{x=L} = 0\right)$



We know our fundamental differential fin equation remains the same, so we will have the same general solution:

$$\theta(x) = C_1 e^{mx} + C_2 e^{-mx}$$

We can re-write this as:

$$\theta(x) = C_1 \cosh(mx) + C_2 \sinh(mx)$$

Our new B.C.'s are:

1) $\theta(x=L) = \theta_b$; 2) $-k \frac{\partial \theta}{\partial x} \Big|_{x=0} = 0$

Remember:

$$\sinh(mx) = \frac{e^{mx} - e^{-mx}}{2}$$

$$\cosh(mx) = \frac{e^{mx} + e^{-mx}}{2}$$

Back substituting our B.C.'s

$$\frac{\partial \theta}{\partial x} = C_1 m \sinh(mx) + C_2 m \cosh(mx)$$

$$\left. \frac{\partial \theta}{\partial x} \right|_{x=0} = C_1 m \underbrace{\sinh(0)}_0 + C_2 m \underbrace{\cosh(0)}_1 = 0$$

$$\boxed{C_2 = 0}$$

Applying our first B.C.

$$\theta(x=L) = \theta_b = C_1 \cosh(mL)$$

$$\boxed{C_1 = \frac{\theta_b}{\cosh(mL)}}$$

Back substituting C_1 & C_2 into our solution:

$$\theta(x) = \frac{\theta_b \cosh(mx)}{\cosh(mL)}$$

$$\boxed{\frac{T - T_\infty}{T_b - T_\infty} = \frac{\cosh(mx)}{\cosh(mL)}} \Rightarrow \text{Temperature profile along the fin for insulated tip.}$$

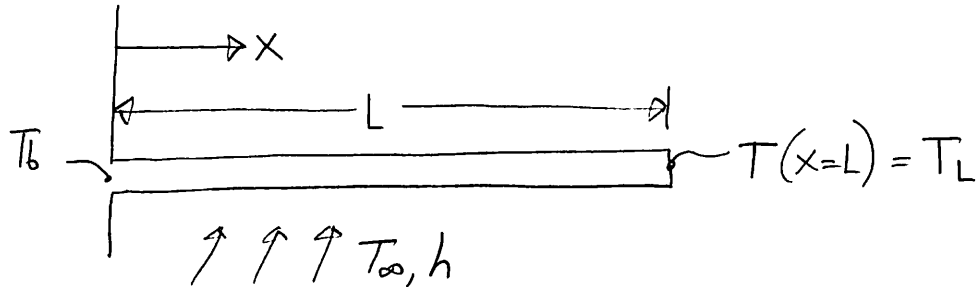
How about heat transfer?

$$Q_{\text{net}} = -kA \left. \frac{\partial \theta}{\partial x} \right|_{x=L} = +kA \theta_b m \frac{\sinh(mL)}{\cosh(mL)} = kA \theta_b \underbrace{\frac{hP \sinh(mL)}{kA \cosh(mL)}}_{\tanh(mL)}$$

$$\boxed{Q_{\text{net}} = \sqrt{kAhP} (T_b - T_\infty) \tanh(mL)}$$

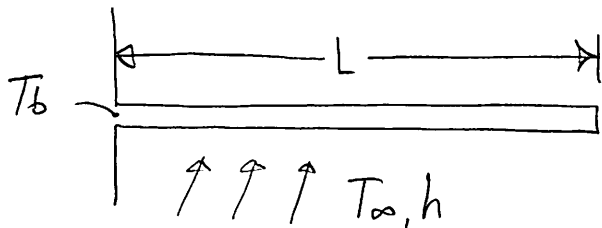
↳ Heat transfer from a fin with an insulated tip.

③ Prescribed Tip Temperature ($T(x=L) = T_L$)



$$\frac{T - T_\infty}{T_b - T_\infty} = \frac{T_L - T_\infty}{T_b - T_\infty} \cdot \frac{\sinh(mx) + \sinh[m(L-x)]}{\sinh(mL)}$$

④ Convection at the Tip ($h(T(x=L) - T_\infty) = -k \frac{\partial T}{\partial x} \Big|_{x=L}$)



$$\frac{T - T_\infty}{T_b - T_\infty} = \frac{\cosh(m(L-x)) + \frac{h}{mk} \sinh(m(L-x))}{\cosh(mL) + \frac{h}{mk} \sinh(mL)}$$

Fin Efficiency

We can define a fin efficiency as:

$$\begin{aligned} \eta_{fin} &= \frac{\text{Actual Heat Transfer with Fin}}{\text{Heat Transfer if } \theta = \theta_b \text{ everywhere on Fin}} \\ &= \frac{q_{actual}}{q_{ideal}} \end{aligned}$$

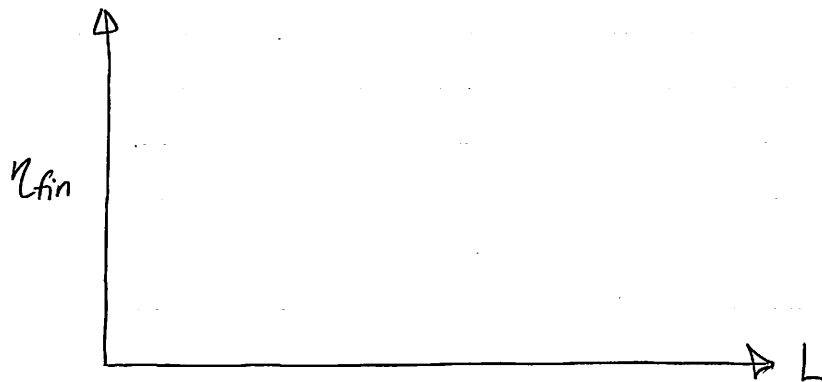
$$q_{ideal} = h(PL)(T_b - T_\infty) \Rightarrow \text{Note this is for the insulated tip case.}$$

$$q_{actual} = \int_0^L hP(T - T_\infty) dx = \sqrt{kAhP} \theta_b \tanh(mL)$$

$$\eta_{fin} = \frac{\sqrt{kAhP} \theta_b \tanh(mL)}{hPL \theta_b} = \underbrace{\frac{\sqrt{kA}}{\sqrt{hP}}}_{1/m} \cdot \frac{\tanh(mL)}{L}$$

$$\boxed{\eta_{fin} = \frac{\tanh(mL)}{mL}}$$

$\Rightarrow$ Insulated Tip Fin Efficiency
For others, see Table 3.5 of Textbook (pg. 168)



Fin Resistance ($R = \Delta T / Q$)

$$R_{fin} = \frac{\theta_b}{q_{actual}} = \frac{T_b - T_\infty}{q_{actual}} = \frac{\theta_b}{\sqrt{kAhP} \tanh(mL) \theta_b} = \frac{1}{hA_{fin} \eta_{fin}}$$

$$\boxed{R_{fin} = \frac{1}{hA_{fin} \eta_{fin}}}$$

$\Rightarrow A_{fin}$ = fin outside area (PL)
 η_{fin} = fin efficiency

Fin Effectiveness (ϵ_{fin})

$$\epsilon_f = \frac{\text{actual heat transfer}}{\text{heat transfer if no fin}} = \frac{q_{ideal} \cdot \eta_{fin}}{q_{ideal} \left(\frac{A}{PL} \right)} \Rightarrow \boxed{\epsilon_f = \eta_{fin} \cdot \frac{PL}{A}}$$